



Aditya Singh

Bachelor of Technology
in Electrical Engineering
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EDUCATION

Degree	Institute/Board	CGPA/Percentage	Year
Bachelor of Technology	Indian Institute of Technology, Ropar	7.07	2020-2024
Senior Secondary	Central Board of Secondary Education	91%	2019-2020
Secondary	Indian Certificate of Secondary Education	89.33%	2017-2018

EXPERIENCE

- Indian Institue of Technology , Ropar** *February 2023 - April 2024*
Worked under Dr.Suman Kumar
– AI edge implimentation.
– Object tracking using AI.
– Image processing testing for edge devices.
- CSIR – Central Scientific Instruments Organisation (CSIO)** *May 2023 - July 2024*
Worked under Dr.Ritesh Kumar
– Object auto-annotation software *Chandigarh*

MAJOR PROJECTS

- AI edge implimentation using TFlite model maker** *January 2023 - April 2023*
– Use TFlite model maker to generate a TFlite model with EfficientDet-0 architecture for image clasification of specific insects with a dataset of around four thousand images.
– Images were collected and annotated by the collaborators.
– We used the TFlite model maker library to custom train an AI under the EfficientDet architecture using those annotated images.
– We tested the model on by running it on a local device for video inputs by breaking video inputs into its constituent frame by frame images.
– Colaborators then further tested the model on an i.MX 8M for edge testing.
- Object auto-annotation software** *May 2023 - July 2023*
– Search for already present options.
– Options turnout to be insufficient.
– Edit in the desired functionality to auto annotate the data on the images fed and then allow manual editing of the annotations into an existing software called Open labeling and calling the new labeling software Semi-labeling.
– Testing and further additions done to the software purely by colaborators.
- Proximity based object tracker** *August 2023 - December 2023*
– Search for already present options.
– Options turnout to be insufficient.
– Creat a proximity based traker from scratch by using a distance threshold to reassign labels to objects.
– Testing the new tracker called Void Track.

TECHNICAL SKILLS

- Programming Languages:** C/C++, Python, Java
- Python Libraries/Skills:** Machine Learning, Neural Networks, pillow (python), OpenCV, MoviePy, TensorFlow, TensorFlow lite, Torch/PyTorch, Glob, Numpy.
- Softwares:** MatLab, Simulink, PSCAD, LTspice, Power World, MS Word, MS PowerPoint, MS Excel.

POSITIONS OF RESPONSIBILITY

- House captain,** Secondary School *Year 2017 - 2018*

KEY COURSES

- Mathematics:** Foundations to Data Science, Multivariable Calculus, Differential Equations , Probability and Stochastic Process, Linear Algebra, Industrial Management, Economics
- Electrical & Electronics:** Digital Circuits, Analog Circuits, Control Engineering, Signal Systems, Internet of things, RF systems, Circuit Theory, Communication Systems, Engineering Electro-Magnetics, Electro-Mechanics , Power systems, Power Electronics, High Voltage Power Equipment
- Physics:** Laser Physics, Physics and aplications of Nanomaterials